

Li Shuai (李帅)

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PROFILE

- LNG trading analyst: 4 years on physical and paper LNG (Centrica, Woodside) and 5 years in energy derivatives at ENGIE (oil, gas, power, coal)
- Strong quantitative background (NUS MFE) with proven ability to build curve, scenario and pricing tools, evaluate contract flexibility and optionality, and turn market views into trade ideas with defined risk/return
- Experienced in cross-functional coordination with Shipping, Risk, Operations, and Contracts to enhance trading accuracy and deal execution, with proficiency in advanced Excel, Python for modeling, and analytics

EDUCATION

National University of Singapore (NUS)

- Aug 22- Dec 24
- Master of Financial Engineering (part-time); focused on option pricing, stochastic calculus, and applying machine learning techniques to model financial risks and optimize trading strategies
 - Implemented Kirk's approximation and change-of-numeraire methods for spread option pricing in Final Year Project: Application of Spread Option in LNG Geographical Arbitrage
- Aug 14- Dec 17
- Bachelor of Business Administration; Double Specialization in Finance and Supply Chain Management Honors with Distinction, Dean's Lister (top 5%)

Victoria Junior College

- Jan 13- Dec 14
- GCE 'A' Levels; Awarded distinctions in Chemistry, Mathematics, Chinese, Physics and Project Work

WORK EXPERIENCE

Jul 23 – Present

Centrica LNG, Singapore

Senior LNG Trading Analyst

- Cover counterparties across spot and term deals: relationship management, negotiation, MSPA review
- Produce short- to long-term analysis of term structure, volatility and market trends for desk positioning
- Build and maintain Python/Excel pricing, scenario, and curve construction frameworks used directly by LNG traders to support commercial decision-making
- Value embedded optionality in term and spot deals (destination flexibility, volume tolerance, cancellation rights) with spread-option and stochastic models; this valuation work supported 4 executed long-term deals (~4 mtpa combined, 10-15 year tenors)
- Mark paper positions to market against close-of-business curves and explain daily P&L moves
- Support traders on paper trade ideas: time spreads, inter-hub spreads, JKM/TTF options
- Run a daily Platts JKM MOC window analysis on both physical and paper transactions (bids, offers and trades against the TTF move) to explain the assessment that JKM paper settles on
- Track vessel movements and Atlantic-Pacific cargo diversions (Kpler) to flag arbitrage openings early
- Monitor in/out-of-the-money contract valuation; automated the desk's portfolio optimization dashboards

Jul 22 – Jul 23

Woodside Energy Trading Singapore Pte. Ltd., Singapore

LNG Trading Analyst

- Conducted market analysis and formulated timely & forward-looking views of gas market developments impacting LNG trading decisions
- Delivered trading and optimization idea write-ups with scenario backtests and risk/return summaries
- Automated daily P&L and exposure analytics across portfolios; implemented reconciliation that reduced P&L breaks and accelerated end-of-day sign-off; reduced manual process by 30%
- Coordinated with Market Risk / Shipping / Ops to align pricing assumptions and delivery optionality; reduced pricing-to-booking discrepancies throughout trade lifecycle
- Built an ADP planning tool and semi-automated nomination and scheduling deadline tracking
- Performed post-deal analysis and reconciliation on P&L against pre-deal shipping economics

Feb 17 – Apr22
(May19 – Apr22)

ENGIE Global Markets, Singapore

Sales and Trading – Commodity Derivatives

- Quoted APAC clients on energy derivatives; developed valuation methods for common options / swaps and improved pricing consistency via standardized inputs and Greeks checks (Oil, Gas, Power, and Coal)
- Worked closely with research team and recommended strategies to clients based on physical needs
- Expanded business via actively searching new clients and generating more deals with existing clients

(Feb17 - May19)

Trade Support Analyst

- Managed trade lifecycle (booking, confirmations, settlements) for futures/options
- Led automation projects by implementing new systems and proposing automation plans
- Enhanced workflow and contributed to projects such as migrating portfolios, managing exotic products

PROJECTS & RESEARCH

- Apr 26 – Present *Event-Driven Algo Trading System (Python, IBKR API)*
- Built an event-driven engine (market → signal → order → fill) that runs the same strategy code in backtest and live IBKR paper trading; every event persisted to SQLite, with a Streamlit monitoring dashboard
 - Risk layer: fixed-risk and ATR volatility sizing, per-symbol and gross-leverage exposure caps, kill switch, order retry and connection heartbeat
 - Screened 20+ technical strategies on Nasdaq-100 stocks and ETFs against buy-and-hold; the screen diagnoses why each strategy fails (over-trading, no edge, weak payoff), so weak ideas are dropped before capital is at risk
 - Research tools: triple-barrier labeling and meta-labeling with purged cross-validation, hierarchical risk parity allocation, CFTC COT positioning and fundamentals filters
- Mar 26 – Present *Personal Portfolio and Risk Dashboard (FastAPI, React)*
- Imports IBKR statements, tracks equity and option positions, and computes position- and portfolio-level Greeks (Black-Scholes) with signal alerts on each price refresh
- Aug 26 – Present *Technical writing: Quant & ML Notes - [lishuai8888.github.io/Personal-Website/learning](https://github.com/lishuai8888/Personal-Website/learning)*
- Published 12 revision notes built from my own code and coursework: derivations worked by hand, each note marked for what is settled and what is still open
 - Topics: machine learning and statistics for trading, model validation, and market microstructure

AWARDS

- NUS CFA Scholarship
- MOE SM1 Program Scholarship
- Gold Award in Singapore Mathematics Competition

SKILLS

- Technical: Python (pandas, NumPy, SciPy), Excel (VBA), Matlab, R, Power BI, Tableau
- Data & platforms: Bloomberg, Platts, ICE, Kpler, Dataloy, Spark, Claude API
- Languages: English and Mandarin (fluent, written and spoken)
- Interests: Archery (NUS team), badminton